Uncertainty from the Small to the Large*†

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October 11, 2021

Abstract

Related decisions are often observed in isolation without direct measurement of correlation in beliefs across state spaces or complementarity in tastes across prize spaces. We introduce a novel model with two decision problems with distinct states and prizes, which we call small worlds, without observation of bets that are contingent on the realization of both worlds. We characterize an appropriate version of subjective expected utility, where choices are made as if there is a joint distribution over the product of the state spaces and a joint utility index over pairs of prizes from both prize spaces. Turning to identification, the joint utility index over pairs of prizes and the marginal belief over each small world is identified, but the uniqueness of the joint distribution is more subtle. If the utility index is separable across prize spaces, then the correlation across state spaces is unidentified; but if there is any complementarity across prizes, then the joint distribution is exactly identified.

Keywords: Uncertainty, Subjective Expected Utility

JEL Codes: D80, D81, D84

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*Declarations of interest: none.

†We thank Haluk Ergin, Larry Karp, Chris Shannon and seminar participants at University of California, Berkeley for helpful feedback.

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1 Introduction

Many environments involve multiple separate but possibly complementary choices across different varieties of contingencies and outcomes. For example, an investor might consider buying shares of equity in Alibaba or JD, the two largest Chinese e-commerce companies. One could sensibly think these assets are positively correlated, since both participate in the Chinese online retail market, or negatively correlated, since they are competitors with each other. If the derivatives market was rich and included assets that depend on the joint returns of both firms, then having such measurements would make the identification of correlation simple. For example, a relative taste for a derivative that pays if both assets appreciate would immediately reveal a belief that the two assets are positively correlated, while aversion to such a parlay would indicate a belief in their negative correlation. However, derivatives of Chinese equity are very limited, so we cannot observe her preferences for such derivatives since they are not available.[1] When markets are incomplete in this manner, the question we ask is whether the investor's joint distribution can be inferred from her investments in the basic assets.

More generally, in many environments where states are correlated, choice in each domain is observed in isolation and without direct instruments to measure believed correlation. Theories of decision-making like subjective expected utility (SEU) typically assume preferences for bets on all joint events are observed by the analyst; that is, they assume preferences for complicated derivatives are available. This paper provides a model that takes the separation of decisions seriously while also acknowledging the possibility of correlation in states or of complementarity in outcomes across decisions. We keep these domains formally detached from each other; that is, we do not assume the existence of complicated bets across domains. We ask what can be inferred about the grand joint problem from observations of choices in smaller problems.

In a famous discussion, Savage discusses the tension between two proverbs: First, the principle of "Look before you leap" suggests enumerating all contingencies and outcomes before making a decision as demanded in observing preferences over all acts in subjective expected utility. Second, the principle that "You can cross that bridge when you come to it" suggests only the immediately relevant details be considered since modeling all possible contingencies seems practically impossible for the decision maker (DM) and analytically impossible for the analyst. Savage ultimately concludes that expected utility theory is best applied to small worlds since "to cross one's bridges when one comes to them means to attack

[1] This is a real consideration given the relatively strict regulation of Chinese asset markets. Even simple put and call options were only recently introduced on Chinese exchanges in 2015.

relatively simple problems of decision by artificially confining attention to so small a world that the 'Look before you leap' principle can be applied there (Savage (1954, p. 16))."

This paper introduces a model that takes an intermediate perspective between small and large worlds. We take seriously that a small world is a set of states and outcomes that is complete in the sense that all bets that assign different outcomes across different states are considered. That is, within a fixed small world the DM looks at all describable bets before she leaps to a decision. An investor decides how much Alibaba stock to buy based on her assessment of its future profits. On the other hand, we also acknowledge that DMs face multiple choices. We therefore consider multiple small worlds simultaneously. So, when deciding how much Alibaba stock to buy, the investor understands how much JD stock she will also hold. However, we also acknowledge that these choices are often made apart from each other. We therefore keep small worlds isolated in the model, in the sense that bets can only be contingent on the realization of a single small world, but cannot depend on the joint realization of states across worlds. The investor cannot invest in a derivative asset whose return jointly depends on the comovement of Alibaba and JD stock prices. Our main question is: What can be inferred about the grand world only from observations across multiple small worlds? This is a theoretically novel question, and we introduce a new framework for analysis to address it.

Formally, we introduce a novel variation of the classic framework by Anscombe and Aumann (1963) (AA) to consider two worlds (or decision problems), each with its own state space, S or T, and its own outcome space, X_S or X_T. We consider preferences over profiles of acts over small worlds, in this case over objects like (f_S, f_T), where $f_S : S \to X_S$ is a bet on the first small state space that returns prize $f_S(s)$ depending on what state s realizes in the first world, and where $f_T : T \to X_T$ is a similar bet that depends on the state of the second world. However, each small world is isolated from the other. Bets can depend on one state space or the other, but cannot have joint contingencies. For example, a bet that pays off only if state $s \in S$ and state $t \in T$ are jointly realized is disallowed. Depending on perspective, our model can be viewed as either an expansion or a restriction of standard models of subjective uncertainty. On one hand, we expand the original Anscombe–Aumann analysis of a small world since now preferences over the marginal act in one world can depend on the held position in the other world. Our model is more rich than an Anscombe–Aumann model for either small world. On the other hand, we restrict the original Anscombe–Aumann analysis of the grand world since now bets that depend on the entire grand state space are prohibited. Our model is more spare than the Anscombe–Aumann model for the large world. In keeping with the original development of Anscombe and Aumann, we will consider lotteries over acts.

Our first result characterizes a suitable version of subjective expected utility for this novel environment. The original Anscombe–Aumann axioms, including some that invoke the randomization over acts, turn out to still characterize maximization of an expected utility index over some fixed subjective belief. In this case, the utility index is defined over the grand prize space, namely $X_S \times X_T$, and the subjective belief is a joint distribution over the grand state space $S \times T$. This shows that the hypothesis of subjective expected utility has the same testable restrictions whether only small worlds are observed in isolation or the grand problem is directly observed with preferences over complicated bets.

We then study the identification of parameters in the representation. Since our model enriches the Anscombe–Aumann model over a small world, we naturally obtain at least that level of identification: The (marginal) belief on either small state space S or T is uniquely identified. However, many candidate joint distributions can share common marginals. The basic mathematical intuition is that the vector space spanned by pairs of marginal acts like (f_S, f_T) where $f_S : S \to X_S$ and $f_T : T \to X_T$, may not span the space of all joint acts from $S \times T$ to $X_S \times X_T$. That is, $(X_S)^S \times (X_T)^T$ does not generally span $(X_S \times X_T)^{S \times T}$. In particular, this gap in the respective spans means that, while marginal beliefs on each dimension of the state space are identified, the whole joint belief may be ambiguous. We show that the precision of mixed moments like correlation is bang-bang. There are two possible cases: In one case, nothing beyond the marginals is known and any subjective belief with those marginals will represent the same behavior; in the other case, the entire subjective belief is known exactly. Importantly, there is no nontrivial partial identification. Which case obtains depends on the structure of the utility index over joint prizes. If there is no interaction between the two prize spaces, that is, if the utility index is additively separable across X_S and X_T, then only the marginals and nothing else about the distribution can be identified. However, if there is any interaction, that is, any incidence of complementarity or substitution, then the entire joint distribution on the large state space $S \times T$ is exactly identified. As an application, we consider the special case where X_S and X_T are money lotteries. In this case, additive separability is shown to be equivalent to risk neutrality. If there are two spaces of investment assets, our main result shows that the correlation in these assets can be exactly identified if there is any diversification benefit implied by risk aversion, but otherwise a risk-neutral investor's believed correlations will be unidentified.

Finally, we leverage the model's product structure to characterize statistical independence across small worlds. We introduce a primitive condition on preferences that guarantees a subjective expected utility representation with a statistically independent belief.

We are not the first to consider formal models of small worlds. In fact, the mentioned discussion in Savage's book is one of the most quoted parts of the whole treatise. Chew and Sagi (2008) consider the problem of identifying small worlds from a larger world. In particular, they identify coarser sub-algebras of the state space where the DM maintains probabilistic sophistication. Their aim is quite different from ours. They use departures of expected utility to endogenize what is a small world. In contrast, we take the small worlds as given and arrive at what looks like a classical expected utility representation over the grand state space.

A paper whose formal model is closer to ours is the work of Ergin and Gul (2009). They also use a product state space, but their aim is to identify issue preference, where the DM prefers to bet on one dimension rather than another. Like the prior reference, they also use departures from probabilistic sophistication to identify the issue preference. The resulting representation would violate a condition we require later, which is essentially indifference to which issue resolved. That is, their theory is designed to detect departures from reduction of compound lotteries, while our model requires no such departures. Similarly, Nau (2006) also considered a product state space to study issue preference via non-reduction of compound lotteries as a model for ambiguity aversion. However, both these papers consider preferences over all acts on the product state space, and do not have separate prize spaces for each issue or constrain the measurability of bets across issues.[2] So our objective, of identifying the subjective probability over the product space, is delivered via an immediate application of known theorems when all acts are compared by the DM.

2 Model

We consider two finite state spaces S, T with $S = \{s_1, s_2, ..., s_N\}, T = \{t_1, t_2, ..., t_M\}$.[3] A complete description of the world specifies the realizations of both S and T, so the grand state space is the product space $S \times T$. Each small state space has an accompanying outcome space X_S or X_T. We employ the Anscombe–Aumann structure and assume each outcome space X_S or X_T is a simplex of simple lotteries, that is, lotteries with finite support, over a deterministic prize space Z_S or Z_T. Then $X_S = \triangle Z_S$ and $X_T = \triangle Z_T$.[4] Let δ_z denote a degenerate lottery concentrated at point z.

Let $\mathcal{F}_S = (X_S)^S$ and $\mathcal{F}_T = (X_T)^T$. That is, $\mathcal{F}_S$ is the set of Anscombe–Aumann acts from state space S to mixture space X_S, and similarly $\mathcal{F}_T$ is the set of all acts from T to X_T . Each problem is a well-defined Anscombe–Aumann domain, but by itself either $\mathcal{F}_S$ or $\mathcal{F}_T$ is an incomplete description of the DM's total problem because it ignores the other choice.

[2]Ergin and Gul (2009) use a purely subjective model, in the sense that they use a Savage primitive and eschew all objective randomization, while we use two layers of explicit randomization in our model.

[3]The limitation to two state spaces is mainly for simplicity and almost all our results generalize to any finite number of worlds.

[4]Throughout the paper, we will use $\triangle Y$ to denote the set of simple lotteries over the set Y.

So we consider the space $\mathcal{F} = \mathcal{F}_S \times \mathcal{F}_T$. That is, the DM holds two positions, one small act $f_S : S \to X_S$ that determines the outcome in Z_S depending on the realization of state $s \in S$ and another similar small act $f_T : T \to X_T$.

We emphasize that $\mathcal{F}$ is not a standard domain of subjective uncertainty in the tradition of Savage or Anscombe–Aumann. The grand state space $S \times T$ is like a usual state space, albeit with some additional structure. The grand prize space is $Z_S \times Z_T$. A standard Anscombe–Aumann model would consider all acts from (grand) states to lotteries over (grand) consequences, that is, any function $f : S \times T \to \triangle(Z_S \times Z_T)$. Our domain is a strict subset of this standard model. It includes only grand acts where the first prize $[f(s,t)]_S \in X_S$ depends only on the realization of state $s \in S$, and similarly the second prize $[f(s,t)]_T$ depends only on $t \in T$. Formally speaking, we consider acts where each projection to a prize space is measurable with respect to the associated state space. This excludes bets where the prize in X_S depends on state t. For example, a basic step function that returns one pair (z_S, z_T) at a fixed grand state (s,t) but returns another pair of prizes (z'_S, z'_T) elsewhere at all $(s', t') \neq (s,t)$ is not part of our model. This exclusion is exactly the point: our goal is to understand what characteristics of the DM can be identified from partial information about small worlds, that is, from observing only variation small world by small world. We exclude bets that parlay both dimensions of the state space. The DM may believe that S and T are correlated or experience complementary utility across X_S and X_T, but these features are not measured directly. Our question is what can be inferred when only behavior in separate small worlds is observed.

We not only consider preferences over acts, but over lotteries of acts. Let $\triangle\mathcal{F}$ denote the family of simple lotteries of $\mathcal{F}$. A lottery that yields the act f^k with probability α^k for $k = 1, 2, ..., K$ is denoted as $Q = \sum_{k=1}^{K} \alpha^k f^k \in \triangle\mathcal{F}$. Our basic primitive is a binary relation over $\triangle\mathcal{F}$. The enriched domain eases the analysis by considering comparisons like whether a fixed act (f_S, f_T) is preferred to a coin flip $\frac{1}{2}(f'_S, f'_T) + \frac{1}{2}(f''_S, f''_T)$ over a better act (f'_S, f'_T) and worse act (f''_S, f''_T). Such comparisons allow cardinal interpretation of the differences in utilities across acts. This is in the spirit of the original Anscombe–Aumann model, and has been recently applied by, for example, Seo (2009) to study ambiguity aversion and by Ergin and Sarver (2015) to study menu choice. We exploit the randomization in a manner similar to these papers, to calibrate cardinal differences across acts. Like these papers, our aim is to characterize a utility index over pure acts $\mathcal{F}$.

An act $f \in \mathcal{F}$ is constant if $f_S(s)$ is constant in s and $f_T(t)$ is constant in t. So a constant act consists of two lotteries, one lottery $p_S \in \triangle Z_S$ over the prize realized in Z_S and another lottery $p_T \in \triangle Z_T$ over the prize realized in Z_T. When it engenders no confusion, we will refer to a constant act by these lotteries (p_S, p_T). We assume that the associated randomization devices are statistically independent. This is actually less objective randomization than the full set of joint distributions over prizes that would be required for the standard Ancombe–Aumann model from $S \times T$ to $\triangle(Z_S \times Z_T)$. That is, we do not assume the existence of a randomization device that can produce all correlation structures, but rather assume only a device that can independently randomize over Z_S and over Z_T. For a pair of marginal distributions p_S over Z_S and p_T over Z_T, their product distribution over $Z_S \times Z_T$ is denoted by $\phi(p_S, p_T) = p_S \times p_T \in \triangle Z$ and is defined in the standard manner for all $(z_S, z_T) \in Z_S \times Z_T$ by:

$$[\phi(p_S, p_T)](z_S, z_T) = [p_S \times p_T](z_S, z_T) = p_S(z_S) \cdot p_T(z_T).$$

This construction can be extended pointwise from constant acts to arbitrary acts. A realization (s, t) of the grand state of the world induces lotteries $f_S(s)$ and $f_T(t)$. Fixing an act, each state is associated with a product measure over $Z_S \times Z_T$. Thus any act $f_S, f_T \in \mathcal{F}$ where $f_S : S \to Z_S$ and $f_T : T \to Z_T$ can be naturally lifted to be a function $\phi(f)$ from $S \times T$ to $\phi(\triangle Z_S, \triangle Z_T)$, that is, from the grand state space to the family of product distributions. This is formally executed in the following manner:

$$[\phi(f)(s, t)](z_S, z_T) = [f_S(s)](z_S) \cdot [f_T(t)](z_T)$$

We will assume that the DM evaluates a pair of lotteries $(p_S, p_T) \in \triangle Z_S \times \triangle Z_T$ as if evaluating the lottery $q = \phi(p_S, p_T) \in \triangle Z$. The pair of lotteries (p_S, p_T) can be viewed as a two-stage compound lottery over Z while q defined above is the one shot reduction of this compound lottery.[5] We will state the assumption explicitly later as the Reduction Axiom and this axiom will be discussed in more detail then.[6] We can further extend ϕ to $\triangle\mathcal{F}$ componentwise: $[\phi(Q)](s, t) := \sum_{k=1}^{K} \alpha^k \cdot \phi(f^k)(s, t)$ so $[\phi(Q)](s, t)$ is a standard mixture of $\phi(f^k)(s, t)$ and is identified as an element in $\triangle Z$. Figure 1 summarizes the discussion of the ϕ operator.

Our primary object of interest is a preference $\succeq$ over the domain $\mathcal{F}$. Like most models of choices under risk and uncertainty, it will be convenient to introduce mixtures, formally convex combinations, of lotteries and acts. The set of constant acts is identified with the space of product measures on $Z_S \times Z_T$. Unfortunately, the family of product measures is not convex. For example, the half-half mixture of $\delta_{(z_S, z_T)}$ and $\delta_{(z'_S, z'_T)}$ is not in $\phi(\triangle Z_S \times \triangle Z_T)$

[5]However, it is important to distinguish the stages of uncertainty here, where one stage is the realization of both states and the next stage is the realization of the resulting lottery, from the two dimensions of uncertainty S and T, whose ordering is irrelevant throughout the paper. We thank an associate editor for suggesting this clarification.

[6]In the pure risk setting, see Segal (1990) for further discussions.

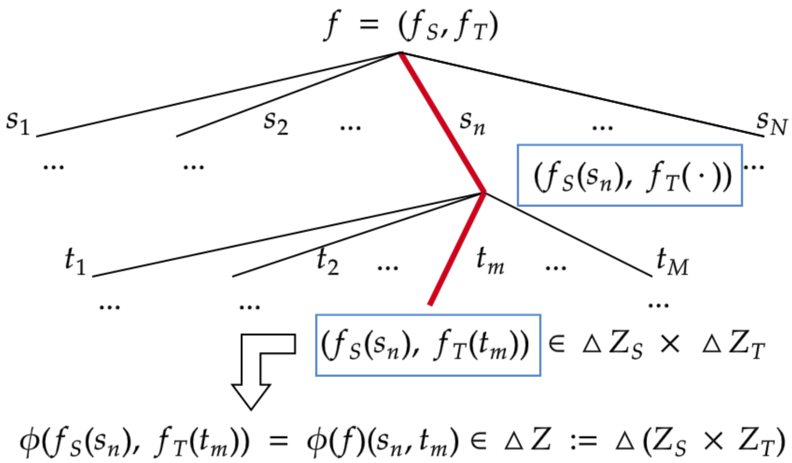

Figure 1: Using the ϕ operator to reduce compound lotteries

whenever $z_S \neq z'_S, z_T \neq z'_T$. Introducing lotteries over $\mathcal{F}$ convexifies the set of constant acts. One way to see this is to observe that any degenerate distribution $\delta_{(z_S,z_T)} \in \triangle(Z_S \times Z_T)$ concentrated on the grand prize (z_S, z_T) is a product measure. These mass points are the extrema of the simplex of lotteries and span $\triangle(Z_S \times Z_T)$. So the convex hull of the family of product measures over joint consequences is the entire $\triangle(Z_S \times Z_T)$. The lack of convexity presents an immediate obstacle to analysis since almost all approaches to expected utility rely heavily on duality approaches that require convexity of domain. Without randomization, the set of constant acts is not even convex. This further explains our use of randomization in our domain $\triangle\mathcal{F}$: Not only is this randomization useful in the usual manner of calibrating sharper information about preferences, but here it serves a more fundamental modeling purpose in closing the consequence space under mixtures.

In contrast to our work, almost all other models of subjective decision-making follow Anscombe and Aumann (1963) and consider a convex preference domain like $\tilde{\mathcal{F}} = \{\tilde{f} : S \times T \to \triangle(Z_S \times Z_T)\}$, instead of $\mathcal{F}$ [7]. Since $\tilde{\mathcal{F}}$ contains all possible mappings from the grand state space to the grand outcome space, we can call it a grand set of acts. As we will show, inferring the DM's subjective belief over the grand state space from our domain $\triangle\mathcal{F}$ is not always possible. This is in contrast to the standard model where observing preferences over $\tilde{\mathcal{F}}$ identifies the entire belief. This difference in identification demonstrates that there is an important modeling distance between the set of grand acts $\tilde{\mathcal{F}}$ and our domain of $\triangle\mathcal{F}$.

[7]Following Anscombe and Aumann (1963), the mixture of two acts in $\tilde{\mathcal{F}}$ can be identified as an act in $\tilde{\mathcal{F}}$, under the reduction of compound lotteries.

3 Subjective Expected Utility with Unique Marginal Beliefs

We first characterize an appropriate version of subjective expected utility for our framework. While many of the conditions will feel familiar from their application in the standard Anscombe–Aumann domain, we caution that their interpretation is slightly different for our more limited domain.

Axiom 1. (Non-triviality)
There exist $z_T \in Z_T, z_S, z'_S \in Z_S$ such that $\delta_{(z_S,z_T)} \succ \delta_{(z'_S,z_T)}$; there exist $z_S \in Z_S, z_T, z'_T \in Z_T$ such that $\delta_{(z_S,z_T)} \succ \delta_{(z_S,z'_T)}$.

This assumption guarantees that each small world is relevant. It requires some strict preference for each dimension, so is a bit stronger than the standard assumption that requires only one incidence of strict preference somewhere.

Axiom 2. (Weak Order)
$\succeq$ is complete and transitive.

This is a standard requirement for rational choice.

Axiom 3. (Continuity)
For any $P \succ Q \succ L$, where $P, Q, L \in \triangle\mathcal{F}$, there exist $\alpha, \beta \in (0,1)$ such that $\alpha P + (1-\alpha)L \succ Q \succ \beta P + (1-\beta)L$.

We invoke a weak Archimedean version of continuity. The stronger topological notion of continuity is also implied by the representation, but Archimedean continuity is sufficient.

Axiom 4. (Monotonicity)
If $P(s,t) \succeq Q(s,t)$ (considered as constant acts) for all s, t, then $P \succeq Q$.

This condition guarantees that the utility for a grand prize pair (z_S, z_T) is invariant to the grand state in which it is obtained, namely state-independence.

Axiom 5. (Independence)
For all $P, Q, L \in \triangle\mathcal{F}$ and all $\alpha \in (0,1)$, $P \succeq Q \iff \alpha P + (1-\alpha)L \succeq \alpha Q + (1-\alpha)L$.

This standard condition maintains its interpretation in our model. It guarantees a linear representation over the space of lotteries over acts. So then there exists a linear expected-utility representation $V : \triangle\mathcal{F} \to \mathbb{R}$ over lotteries of acts. If $Q = \sum_{k=1}^{K} \alpha^k (f_S^k, f_T^k)$ is a lottery over acts $(f_S^1, f_T^1), (f_S^2, f_T^2), \ldots, (f_S^K, f_T^K)$ with weights $\alpha^1, \alpha^2, \ldots, \alpha^K$, then its expected utility is $V(Q) = \sum_{k=1}^{K} \alpha^k W(f_S^k, f_T^k)$. Our main representation sharpens the structure of W, that is, it is a formula for the value of degenerate lotteries over acts.

The final axiom is somewhat less common than the others in modern versions of the Anscombe–Aumann model, and invokes the structure of our expanded domain that includes lotteries over acts.

Axiom 6. (Reduction)
For any $P, Q \in \triangle\mathcal{F}, \phi(Q) = \phi(P) \implies Q \sim P$.

This condition requires the DM to care only about the reduced distribution of prizes returned in each state of the world. For example, consider $Q = \frac{1}{2}f + \frac{1}{2}g, P = l$, where f, g, l are all constant acts and $f = \delta_{(z_S^1, z_T)}, g = \delta_{(z_S^2, z_T)}, l = (p_S, \delta_{z_T})$ with $p_S(z_S^1) = p_S(z_S^2) = \frac{1}{2}$. Clearly, P, Q are different elements in $\triangle\mathcal{F}$ but $\phi(P) = \phi(Q)$. The Reduction Axiom simply requires indifference between P and Q. The standard interpretation is the reduction of compound lotteries, or equivalently neutrality towards the timing of uncertainty. A lottery over Anscombe-Aumann acts is a compound lottery: first, an act (f_S, f_T) is realized from the lottery over acts, then second a state (s, t) is realized and a lottery $\phi(f_S(s), f_T(t))$ over prizes is paid. Instead, suppose we first realized a state (s, t), then second draw a lottery over which act's payoff is realized. The Reduction Axiom asserts the DM's indifference between these two orders for realizing uncertainty.

It is important to note that the reduction here is between two specific stages of grand uncertainty: the realization of the grand state (s, t) and the realization of the implied prize lottery $f(s, t)$. Our Reduction Axiom is the standard one from the original Anscombe–Aumann model, and would be sensible even if the state space did not have the product structure $S \times T$. That is, it is important to distinguish the reduction of these spaces from the reduction of the small world S and the small world T. The Reduction Axiom has no bite across those spaces. The agent reduces uncertainty across grand states and grand prizes, but the condition is silent on how uncertainty is reduced across small components of the grand state space.

Nearly all recent applications of the Anscombe–Aumann framework drop randomization over acts, so take no position on compounding. However, restrictions that are very similar to the reduction axiom appear in the original treatment by Anscombe and Aumann that explicitly considers lotteries over acts. That is, if S is a state space and X is a mixture space, then the original Anscombe–Aumann domain was actually $\triangle(X^S)$, even though most modern treatments study preferences over X^S. In many applications, the additional randomization provides no benefits because the relevant representation can be characterized and identified with the sparser domain. Here, the richness of randomized acts is crucial in obtaining our results.

One contribution of the following representation theorem is to understand the force of the original Anscombe–Aumann axiom scheme when applied over a product of small worlds, rather than directly over all grand acts. The following result shows that these conditions remain sufficient for subjective expected utility. That is, requiring the axioms of expected utility over larger space of all product acts assumes more than what is required to guarantee an expected utility representation. However, as we will explore in more detail in the sequel, there will be a gap between the uniqueness of the parameters across the two domains. That is, the analytic loss in the Anscombe–Aumann framework of considering only small worlds is not in the testable content of subjective expected utility, but rather in the sharpness of its implicit parameters.

Theorem 1. *(Subjective Expected Utility with Unique Marginal Beliefs)*
$\succeq$ *satisfies Axioms 1-6 if and only if there exist a utility index* $u : Z \to \mathbb{R}$ *and a probability measure* $\mu \in \triangle(S \times T)$ *such that, for* $Q = \sum_k \alpha^k f^k$,

$$V(Q) = \sum_k \alpha^k W(f^k)$$

defines a utility representation of $\succeq$ *where*

$$W(f) = \int_{S\times T} U\Big(\phi\left(f_S^k(s), f_T^k(t)\right)\Big) d\mu(s,t) \tag{1}$$

and

$$U(q) = \int_Z u(z)\, dq(z). \tag{2}$$

Moreover, u *is nonconstant and unique up to positive affine transformation and the marginal beliefs* $\mu_S(s_n) := \sum_m \mu(s_n, t_m), \mu_T(t_m) := \sum_n \mu(s_n, t_m)$ *for any* s_n, t_m *are unique.*[8]

Here, as in the original Anscombe–Aumann domain, the main object of interest is not the expected-utility function $V : \triangle\mathcal{F} \to \mathbb{R}$ over lotteries of acts, but rather the Bernoulli index $W : \mathcal{F} \to \mathbb{R}$ over (degenerate) acts. The axioms guarantee that the cardinal utility for an act is its expected utility $u : Z_S \times Z_T \to \mathbb{R}$ over the joint consequences that is integrated using a subjective product measure $\mu \in \triangle(S \times T)$ over all joint states.

We briefly sketch the proof here. The Reduction Axiom allows us to construct a preference relation $\dot{\succeq}$ over $\phi(\triangle\mathcal{F})$ defined by $\phi(Q)\dot{\succeq}\phi(P)$ if and only if $Q \succeq P$. Next, we apply the von Neumann–Morgenstern Expected Utility Theorem for objective risk to the set of lotteries

[8]That is, if $(u, \mu), (u', \mu')$ both represent $\succeq$, $\mu_S = \mu'_S, \mu_T = \mu'_T$.

over constant acts $\triangle(Z_S \times Z_T)$ and identify the expected-utility function $U : \triangle Z \to \mathbb{R}$. So the utility index u over grand prizes is immediately revealed through the lotteries over constant acts. Note that without convexification this would not be possible, since $\mathcal{F}$ does not include all distributions over $Z = Z_S \times Z_T$ while $\triangle\mathcal{F}$ does contain all such lotteries. Now define a mapping $\pi : \phi(\triangle\mathcal{F}) \to \mathbb{R}^{NM}$ such that the nmth coordinate of $\pi(\phi(Q))$ is given by $U(\phi(Q)(s_n, t_m))$. We verify that $\pi(\phi(\triangle\mathcal{F}))$ is a convex subset of $\mathbb{R}^{NM}$.[9] Next, define the binary relation R over $\pi(\phi(\triangle\mathcal{F})$ by $\pi(\phi(Q))R\pi(\phi(P))$ if and only if $\phi(Q)\dot{\succeq}\phi(P)$. We show R satisfies the axioms required by the Mixture Space Theorem and thus obtain a linear representation. The Mononicity Axiom implies the weights of the linear representations are nonnegative and thus can be normalized as a belief μ.

A key difference between our findings and the canonical Anscombe–Aumann representation is that the DM's subjective belief over the grand state space $S \times T$ is not uniquely identified. Instead, only the marginal beliefs μ_S, μ_T are generally pinned down. The next section will discuss additional structures that will guarantee the uniqueness of the subjective belief. However, it may be useful to now briefly discuss some of the geometry for the lack of identification. The proof proceeds by generating a linear function V on the span of $\pi(\phi(\triangle\mathcal{F}))$. The key issue is that $\pi(\phi(\triangle\mathcal{F}))$ may not span all of $\mathbb{R}^{NM}$. In that case, there are multiple linear extensions of V from the span of $\pi(\phi(\triangle\mathcal{F}))$ to $\mathbb{R}^{NM}$, and any of these extensions also represent the same behavior. It turns out that the existence of a gap between the span and the entire vector space can be verified with a condition on deterministic preferences, namely additive separability.

4 Additively Separable Preference and Unique Belief

This sections studies preferences that are additively separable across prize spaces Z_S and Z_T, that is, where the utility index $u : Z_S \times Z_T \to \mathbb{R}$ of representation (1) can be decomposed as $u(z_S, z_T) = u_S(z_S) + u_T(z_T)$ for some $u_S : Z_S \to \mathbb{R}$ and $u_T : Z_T \to \mathbb{R}$. The standard interpretation of additive separability is the absence of complementarity and substitution across prizes. For example, time-separable preferences across streams of consumption in period 1 Z_1 and in period 2 Z_2 are additively separable. We study two distinct approaches to characterizing additively separable preference. The first is a direct approach, imposing a restriction on preferences over constant acts. In the second approach, we impose restrictions on conditional preferences. If the preference over payoffs on one dimension is invariant to the payoffs from the other dimension, then preferences are additively separable. While such hexagon-like conditions are familiar for characterizing additive separability, our main insight

[9]Note that $\pi(\phi(\mathcal{F}))$ is not necessarily convex, which motivates our convexification of the domain.

in this section is that a violation of additive separability, that is, some incidence of nontrivial complementary or substitution across prizes, is necessary and sufficient to identify the joint belief. Conversely, if preferences are additively separable, then beliefs are identified only up to marginals.

4.1 Direct Characterization of Additive Separability

Our first characterization of additively separable preferences directly imposes restrictions on preferences over objective risk, that is, over lotteries of constant acts. Additive separability has a long tradition in the theory of measurement and we follow the treatment for expected utility theory by Fishburn (1982). Let $Q \in \triangle\mathcal{F}$ be a lottery over constant acts and then $\phi(Q)$ can be viewed as a lottery in $\triangle Z$. For any element $z_S \in Z_S$, we write $\phi(Q)_S(z_S) = \sum_{z_T \in Z_T} \phi(Q)(z_S, z_T)$. So $\phi(Q)_S(z_S)$ is the marginal probability that the DM obtains prize pairs containing z_S in the first dimension (and $\phi(Q)_T(z_T)$ can be defined similarly). The main axiom in Fishburn (1982) requires the DM to be indifferent between two lotteries in $\triangle Z$ if their marginal probabilities over Z_S, Z_T are the same, which is translated to our context as follows.

Axiom 7. (Additive Separability)
Let $P, Q \in \triangle\mathcal{F}$ be lotteries over constant acts. If $\phi(P)_S(z_S) = \phi(Q)_S(z_S)$ for all $z_S \in Z_S$ and $\phi(P)_T(z_T) = \phi(Q)_T(z_T)$ for all $z_T \in Z_T$, then $P \sim Q$.

The Additive Separability Axiom requires that for a joint distribution on $Z_S \times Z_T$, its marginal lotteries over Z_S and Z_T are sufficient statistics for the expected utility of that lottery. This indifference to mixed moments characterizes additive separability across small prize spaces. In addition, if there is indifference to mixed moments of a lottery over the grand prize space, then the analyst cannot estimate the mixed moments of the subjective belief over the grand state space $S \times T$. In fact, *any* subjective belief with the same marginals will generate the same observed behavior.

Theorem 2. *(SEU with Additively Separable Utility)*
$\succeq$ satisfies Axioms 1-7 if and only if it is represented (u, μ) as in (1) where u is additively separable, that is, $u(z_S, z_T) = u_S(z_S) + u_T(z_T)$.

Moreover, let (u_S, u_T, μ) represent $\succsim$. Then (u'_S, u'_T, μ') also represents $\succsim$ if and only if:

1. *$u'_S = au_S + b$ and $u'_T = au_T + b$ for fixed $a > 0$ and $b \in \mathbb{R}$, and*

2. *$\mu'_S = \mu_S$ and $\mu'_T = \mu_T$.*

Suppose that the utility index u is additively separable, or equivalently that $\succsim$ satisfies the Additive Separability Axiom. Then the two dimensions of the utility index, u_S and u_T, are identified up to joint positive affine transformation and the belief μ is identified up to marginals. That is, the correlation structure of μ is completely unidentified and the analyst cannot discriminate between even perfect positive or negative correlation.

The intuition behind the lack of uniqueness in Theorem 2 is straightforward. While more details are in the proof, if u is additively separable, then the representation (1) can be decomposed into two parts that each only depends on one dimension. In other words, the whole decision problem can be decomposed into two independent sub-problems and therefore any correlation across S, T is irrelevant. This observation also motivates our second approach to characterizing additive separability, which appeals to these conditional preferences.

4.2 Conditional Preference and Additive Separability

For a fixed $f_S \in \mathcal{F}_S$, we can define the restricted domain $\mathcal{F} \mid f_S := \{g \in \mathcal{F} : g_S(s) = f_S(s) \text{ for all } s \in S\}$. For a preference $\succeq$ over $\triangle\mathcal{F}$, its restriction over $\mathcal{F} \mid f_S$ is the conditional preference, which is denoted as $\succeq_{f_S}$. There is clearly a natural isomorphism between $\mathcal{F} \mid f_S$ and $\mathcal{F}_T$ and thus a conditional preference $\succeq_{f_s}$ induces a preference relation over $\mathcal{F}_T$. Similarly, for a fixed $f_T \in \mathcal{F}_T$, the restriction of $\succeq$ over $\mathcal{F} \mid f_T$ is the conditional preference $\succeq_{f_T}$, which induces a preference relation over $\mathcal{F}_S$.

In general, if $f_S \neq g_S$, then the conditional preferences $\succeq_{f_S}, \succeq_{g_S}$ over $\mathcal{F}_T$ may diverge. This simply reflects the interaction of prizes. A natural question that arises is when these conditional preferences are the same for different f_S's. It turns out that if the DM's preference $\succeq$ over $\triangle\mathcal{F}$ is represented by an SEU representation (1), then these conditional preferences agree if and only if u is additively separable. That is, if preferences over bets on state space T are independent of the bets held on state space S, then preferences over joint prizes exhibit no interaction. This invariance of conditional preferences is formally summarized by the following two axioms.

Axiom 8. (Identical Conditional Preferences S)
For any $f_S, g_S \in \mathcal{F}_S$, the marginal preferences $\succeq_{f_S}, \succeq_{g_S}$ over $\mathcal{F}_T$ agree.

Axiom 9. (Identical Conditional Preferences T)
For any $f_T, g_T \in \mathcal{F}_T$, the marginal preferences $\succeq_{f_T}, \succeq_{g_T}$ over $\mathcal{F}_S$ agree.

Theorem 3. *(Conditional Preferences and Additive Separability Under SEU)*
Suppose the preference $\succeq$ over $\triangle\mathcal{F}$ is represented by SEU (1), then the following statements are equivalent:

1. $\succeq$ satisfies the Additive Separability Axiom.

2. u is additively separable as in Theorem 2.

3. $\succeq$ satisfies the Identical Conditional Preference S.

4. $\succeq$ satisfies the Identical Conditional Preference T.

Theorem 3 provides an alternative characterization to Theorem 2. The equivalence between the three axioms (under the SEU) is somewhat surprising because their interpretations are quite different. In particular, the Additive Separability Axiom directly imposes restrictions on the preferences without uncertainty. While the Identical Conditional Preference S (T) Axiom requires the DM's preference over payoffs from one dimension to be unaffected by the background uncertainty in the other dimension.

4.3 Uniqueness of Belief

Theorem 2 shows that additive separability implies that the belief μ in the SEU representation can only be identified up to its marginals μ_S, μ_T. We now show that the converse of this statement is also true: if the preference is *not* additively separable, then the belief μ in the SEU is unique. Note that any single violation of additive separability is sufficient, so the two theorems cover all cases. Therefore the only possible cases are either that only the marginals are identified and nothing else, or that the entire distribution is identified. There are no intermediate partial identification regions.

Theorem 4. *(Uniqueness of Belief)*
Suppose the preference $\succeq$ over $\triangle\mathcal{F}$ has an SEU representation (1). Then the belief μ is unique if $\succeq$ is not additively separable.

To sketch the proof of Theorem 4, we introduce some new notations. For any $s \in S, t \in T$, define $\mathcal{F}^{st} =: \{f \in \mathcal{F} : f_S(s') = f_S(s''), f_T(t') = f_T(t'') \text{ for all } s', s'' \neq s,\ t', t'' \neq t\}$. An act f in $\mathcal{F}^{st}$ with $f_S(s) = p_S, f_S(s') = p'_S$ for all $s' \neq s$, $f_T(t) = p_T, f_T(t') = p'_T$ for all $t' \neq t$ can be rewritten as $((p_S, p'_S)^s, (p_T, p'_T)^t)$ for convenience. Then $\succeq$ over the restricted domain $\mathcal{F}^{st} \subset \mathcal{F}$ is evaluated by:

$$
\begin{aligned}
V(f) =& \mu(s,t)U\Big(\phi(p_S, p_T)\Big) + \mu(S \setminus s, t)U\Big(\phi(p'_S, p_T)\Big) \\
&+ \mu(s, T \setminus t)U\Big(\phi(p_S, p'_T)\Big) + \mu(S \setminus s, T \setminus t)U\Big(\phi(p'_S, p'_T)\Big)
\end{aligned}
$$

Following a similar strategy as in the proof for Theorem 1, we define a preference relation R over $\pi(\phi(\triangle\mathcal{F}^{st}))$ such that $\pi(\phi(Q))R\pi(\phi(P))$ if and only if $Q \succeq P$ for any $P, Q \in \triangle\mathcal{F}^{st}$.

Note that each element $\pi(\phi(Q)) \in \pi(\phi(\triangle\mathcal{F}^{st}))$ can be identified as an element in $\mathbb{R}^4$ because by construction $\pi^{nm}(\phi(Q)) = \pi^{ij}(\phi(Q))$ if $s_n, s_i \in S \setminus s, t_m, t_j \in T \setminus t$. For example, for the f above, $\pi(\phi(f))$ can be identified by

$$\left(U\Big(\phi(p_S, p_T)\Big), U\Big(\phi(p'_S, p_T)\Big), U\Big(\phi(p_S, p'_T)\Big), U\Big(\phi(p'_S, p'_T)\Big)\right) \in \mathbb{R}^4$$

Thus V is a linear function representing a preference relation R over $\pi(\phi(\triangle\mathcal{F}^{st}))$ [10], which is identified as a subset $\mathcal{U}$ of $\mathbb{R}^4$. If we can show the linear representation V above is unique then we are done, since $\mu(s,t)$ is then unique for arbitrary s,t. By Lemma 1, it is sufficient to show that $\mathcal{U}$ is convex, contains the zero vector, and spans $\mathbb{R}^4$. The convexity is free because we convexified the domain. Because U in the SEU representation is unique up to positive affine transformation, we can choose a U such that $\mathcal{U}$ contains the zero vector in $\mathbb{R}^4$ by considering a constant act and shifting uniformly to the origin. It remains to prove $\mathcal{U}$ spans $\mathbb{R}^4$, and the details of this proof are in the appendix.

4.4 Application: Monetary Payoffs

We apply these results to the special but important case where the outcome spaces are monetary payoffs where $Z_S = Z_T = \mathbb{R}$. This can be interpreted as investment returns from two different assets, where the return of each asset depends only on one dimension of uncertainty. Then the DM only cares about the total monetary payoff summed across the two investments. We therefore specialize the main representation (1) so that there exists $v : \mathbb{R} \to \mathbb{R}$ such that $u(z_S, z_T) = v(z_S + z_T)$ for all $z_S \in \mathbb{R}, z_T \in \mathbb{R}$.

Let $\phi^* : \triangle\mathbb{R} \times \triangle\mathbb{R} \to \triangle\mathbb{R}$ be the convolution operator that outputs the sum $\phi^*(p_S, p_T)$ of two random variables p_S, p_T in the standard manner by counting the number of ways that a sum z of payoffs can be realized across the two monetary bets, that is,

$$\phi^*(p_S, p_T)(z) = \sum_{z_T + z_S = z} p_S(z_S) \cdot p_T(z_T).$$

We can apply the convolution operator to a particular lottery of acts $Q = \sum_{k=1}^{K} \alpha^k f^k \in \triangle\mathcal{F}$ as follows:

$$\phi^*(Q)(s,t) = \sum_{k=1}^{K} \alpha^k \phi^* \left(f_S^k(s), f_T^k(t)\right).$$

Axiom 10. (Monetary Payoffs)

[10] Recall $\triangle$ is used similarly as before to convexify the domain.

- For any $P, Q \in \triangle\mathcal{F}, \phi^*(Q) = \phi^*(P) \implies Q \sim P$.
- $z_S + z_T > z'_S + z'_T \implies \delta_{(z_S, z_T)} \succ \delta_{(z'_S, z'_T)}$

The Monetary Payoffs axiom has two parts. The first part requires that the DM only cares about the total monetary payoffs and this part is stronger than the Reduction axiom because $\phi(P) = \phi(Q) \implies \phi^*(P) = \phi^*(Q)$. The second part of the axiom simply says more money is strictly better and it is clear the second part implies the Non-triviality axiom.

Theorem 5. *(Monetary SEU)*
$\succeq$ *satisfies Axioms 2–5 and 10 if and only if there exist a strictly increasing utility index over money* $v : \mathbb{R} \to \mathbb{R}$ *and a probability measure* $\mu \in \triangle(S \times T)$ *such that, for* $Q = \sum_k \alpha^k f^k$,

$$V(Q) = \sum_k \alpha^k W(f^k)$$

defines a utility representation where

$$W(f) = \int_{S\times T} U\Big(\phi^*\left(f^k_S(s), f^k_T(t)\right)\Big) d\mu(s,t) \tag{3}$$

and for $q \in \triangle\mathbb{R}$,

$$U(q) = \int_{\mathbb{R}} v(z)\, dq(z).$$

Moreover, v *is unique up to positive affine transformation and the marginal beliefs* μ_S, μ_T *are unique.*

We say v is risk neutral if for any $x, y \in \mathbb{R}$, $\frac{1}{2}v(x) + \frac{1}{2}v(y) = v(\frac{1}{2}x + \frac{1}{2}y)$. Since Axiom 10 (Monetary Payoffs) implies that the DM only cares about the sum of the monetary payoffs, v is risk-neutral if and only if u is additively separable. To see the forward "only if" direction, suppose v is risk-neutral. Let $u_S = u_T = v$. Then $u(z_S, z_T) = v(z_S + z_T) = v(z_S) + v(z_T) = u_S(z_S) + u_T(z_T)$, so u is additively separable. For the converse "if" direction, suppose u is additively separable, $u(z_S, z_T) = u_S(z_S) + u_T(z_T)$. To prove v is risk neutral, it suffices to show $\frac{1}{2}v(x) + \frac{1}{2}v(y) = v(\frac{1}{2}x + \frac{1}{2}y)$ for all $x, y \in \mathbb{R}$. By the definition of v,

$v(x+y) = u(x,y) = u_S(x) + u_T(y)$ for all $z_S, z_T \in \mathbb{R}$. It follows

$$\begin{aligned}
\frac{1}{2}v(x) + \frac{1}{2}v(y) &= \frac{1}{2}\left[u_S\left(\frac{x}{2}\right) + u_T\left(\frac{x}{2}\right)\right] + \frac{1}{2}\left[u_S\left(\frac{y}{2}\right) + u_T\left(\frac{y}{2}\right)\right] \\
&= \frac{1}{2}\left[u_S\left(\frac{x}{2}\right) + u_T\left(\frac{y}{2}\right)\right] + \frac{1}{2}\left[u_S\left(\frac{y}{2}\right) + u_T\left(\frac{x}{2}\right)\right] \\
&= \frac{1}{2}u\left(\frac{x}{2}, \frac{y}{2}\right) + \frac{1}{2}u\left(\frac{y}{2}, \frac{x}{2}\right) \\
&= \frac{1}{2}v\left(\frac{x}{2} + \frac{y}{2}\right) + \frac{1}{2}v\left(\frac{y}{2} + \frac{x}{2}\right) = v\left(\frac{x}{2} + \frac{y}{2}\right)
\end{aligned}$$

We apply Theorem 2 and Theorem 4 to obtain the following result as a proposition.

Proposition 1. *(Belief Identification and Risk Neutrality)*
Suppose $\succeq$ is represented by V as (3), then

- *v is risk neutral $\implies$ if $\mu'_S = \mu_S, \mu'_T = \mu_T$, then (U, μ') also represents $\succeq$.*
- *v is not risk neutral $\implies$ μ is unique.*

To provide intuition for Proposition 1, consider the following investment example of how to identify beliefs under risk aversion. Suppose the DM has a monetary SEU representation with a strictly concave utility index over money v. The DM can invest in a stock and in a bond, each of which could have a high or low return. Let $S = \{H, L\}, T = \{h, l\}$, where H and L denotes the high and low stock prices, and h and l denotes analogous bond prices.

Since marginal beliefs are unique, we let $\alpha = \mu_S(H) =: \mu(H,h) + \mu(H,l), \beta = \mu_T(h) =: \mu(H,h) + \mu(L,h)$ and take α, β as given. Assume $\alpha \in (0,1), \beta \in (0,1), \alpha + \beta < 1$ for convenience. Let $\gamma = \mu(H,h)$ and it is clear that if γ is identified, then μ over $S \times T$ will also be identified. We propose a way to recover γ.

Consider an act f such that for any $\epsilon \in \mathbb{R}$,

$$\begin{aligned}
f_S(H) = \frac{1}{2}\epsilon; \quad f_S(L) = -\frac{1}{2}\epsilon \\
f_T(h) = \frac{1}{2}\epsilon; \quad f_T(l) = -\frac{1}{2}\epsilon
\end{aligned}$$

Call f "act ϵ" and the act ϵ can be interpreted as long positions in both stock and bond. The payoff matrix is given in Table 1.

The DM evaluates the act ϵ by:

$$V(\epsilon) = \gamma v(\epsilon) + (\alpha + \beta - 2\gamma)v(0) + (1 - \alpha - \beta + \gamma)v(-\epsilon)$$

		Bond	
		h	l
Stock	H	$\frac{1}{2}\epsilon + \frac{1}{2}\epsilon$	$\frac{1}{2}\epsilon - \frac{1}{2}\epsilon$
	L	$-\frac{1}{2}\epsilon + \frac{1}{2}\epsilon$	$-\frac{1}{2}\epsilon - \frac{1}{2}\epsilon$

Table 1: Payoff Matrix of "act ϵ"

When $\epsilon = 0$, the DM receives \$0 with certainty. Consider the DM's choice between the risky act ϵ and receiving \$0 with certainty. In particular, she prefers the certainty prize \$0 if and only if

$$V(0) \geq V(\epsilon) \iff v(0) \geq \frac{\gamma}{1-\alpha-\beta+2\gamma} v(\epsilon) + \frac{1-\alpha-\beta+\gamma}{1-\alpha-\beta+2\gamma} v(-\epsilon)$$

The RHS is the expected utility from a lottery that yields ϵ with probability $\frac{\gamma}{1-\alpha-\beta+2\gamma}$ and yields $-\epsilon$ otherwise. The expected return from the RHS lottery is equal to $-\frac{1-\alpha-\beta+\gamma}{1-\alpha-\beta+2\gamma}\epsilon$ and the expected return is positive if and only if $\epsilon < 0$. In order to make the risk-averse DM to be indifferent, the expected return from the RHS lottery must be large enough to compensate the DM. Let's assume there exists $\epsilon^* < 0$ such that the act $\epsilon^* \sim \$0$ so \$0 is the certainty equivalence for the act ϵ^*. It follows γ can be solved by the equation $V(0) = V(\epsilon^*)$, which establishes the uniqueness of γ and μ.

The argument above does not work for the case when v is risk neutral. When the DM is risk neutral, she only cares about the expected return of the RHS lottery. Note that the expected return $-\frac{1-\alpha-\beta+\gamma}{1-\alpha-\beta+2\gamma}\epsilon \neq 0$ whenever $\epsilon \neq 0$. Thus we cannot find nonzero ϵ^* such that \$0 is the certainty equivalence for the act ϵ^*.

5 Behavioral Characterization of Statistical Independence

This final section examines how to test whether the DM believes the two state spaces are correlated. Note that independence of belief is a weaker hypothesis than additive separability, since there can be complementarity over prizes without correlation over states. We provide necessary and sufficient conditions for an SEU representation with an independent belief over $S \times T$. The following definition is standard.

Definition 1. (Statistically Independent Belief)
A belief (probability measure) μ over $S \times T$ is **statistically independent** if, for any $s \in S, t \in T$, $\mu(s,t) = \mu_S(s) \cdot \mu_T(t)$, where μ_S, μ_T are the marginal beliefs.

For any $s \in S$ and $f, g \in \mathcal{F}$, we define $f \diamond_s g = h$ such that $h_S(s) = f_S(s), h_S(s') = g_S(s')$ for all $s' \neq s$ and $h_T = g_T$. In other words, $f \diamond_s g = h$ agrees with g except its prize in the S dimension is the same as f if s is realized. Similarly, for any $t \in T$, we define $f \diamond_t g = h$ such that $h_T(t) = f_T(t), h_T(t') = g_T(t')$ for all $t' \neq t$ and $h_S = g_S$. We introduce a subset of the acts $\mathcal{F}$ that leads to payoffs with certainty in S or T dimension, which we call the marginally certain acts.

Definition 2. (Marginally Certain Acts)
Let $\mathcal{F} \mid S := \{f \in \mathcal{F} : f_S(s) = f_S(s')$ for all $s \in S\}$ and $\mathcal{F} \mid T := \{f \in \mathcal{F} : f_T(t) = f_T(t')$ for all $t \in T\}$. We call $\mathcal{F} \mid S$ the $S-$certain acts and $\mathcal{F} \mid T$ the $T-$certain acts.

We can now introduce the behavioral conditions of interest.

Axiom 11. (Statistical Independence S)
For any $s \in S$ and $f, g, f', g' \in \mathcal{F} \mid S$ such that $f_T = g_T, f'_T = g'_T$, if $f \not\sim g$ and $f \diamond_s g \sim \alpha f + (1-\alpha) g$ for some $\alpha \in [0,1]$, then $f' \diamond_s g' \sim \alpha f' + (1-\alpha) g'$.

Axiom 12. (Statistical Independence T)
For any $t \in T$ and $f, g, f', g' \in \mathcal{F} \mid T$ such that $f_S = g_S, f'_S = g'_S$, if $f \not\sim g$ and $f \diamond_t g \sim \alpha f + (1-\alpha) g$ for some $\alpha \in [0,1]$, then $f' \diamond_t g' \sim \alpha f' + (1-\alpha) g'$.

We will explain the axioms in more detail after we introduce the theorem.

Theorem 6. *(SEU with Statistically Independent Beliefs)*
Suppose $\succeq$ has an SEU representation (1), then the following are equivalent:

1. *$\succeq$ has an SEU representation (1) with a statistically independent μ.*
2. *$\succeq$ satisfies the Statistical Independence S Axiom.*
3. *$\succeq$ satisfies the Statistical Independence T Axiom.*

The first statement of Theorem 6 asserts the existence of a statistically independent μ, but says nothing about its uniqueness and allows for the possibility of other representing measures. If μ in the SEU is unique, then it must be statistically independent. If the beliefs are not unique, by Theorem 4, the product measure μ that is generated by its marginals will also rationalize the preference. Therefore, the Statistical Independence S, T axioms are logically weaker than assuming that the preference is additively separable.

We discuss only the Statistical Independence S axiom since the intuition behind Statistical Independence T axiom is symmetric. For simplicity, assume the small state spaces are binary $S = \{s_1, s_2\}, T = \{t_1, t_2\}$. Further assume the small prize spaces are binary

$Z_S = \{0, 1\}, Z_T = \{\$50, \$100\}$, e.g., in the S dimension the prizes are chocolates and in the T dimension the prizes are dollars.

Consider constant acts $f = (1, \$50), g = (0, \$50)$ and assume more is better, so $f \succ g$. Then $h = f \diamond_{s_1} g$ will deliver f if $s_1 \in S$ is realized and deliver g otherwise. Under SEU, the DM does not care whether the probability is generated objectively or by her subjective beliefs. Thus she must be indifferent between h and an objective lottery that yields f with $\mu_S(s_1)$ and g with $1 - \mu_S(s_1)$. So $\mu_S(s_1)$ is a good candidate for the α in the axiom.

Next we consider acts f', g' and $h' = f' \diamond_{s_1} g'$ as follows (see Figure 2):

$$\begin{aligned} f'_S = 1; \quad & f'_T(t_1) = \$50 \quad f'_T(t_2) = \$100 \\ g'_S = 0; \quad & g'_T(t_1) = \$50 \quad g'_T(t_2) = \$100 \\ h'_S(s_1) = 1 \quad h'_S(s_2) = 0; \quad & h'_T(t_1) = \$50 \quad h'_T(t_2) = \$100 \end{aligned}$$

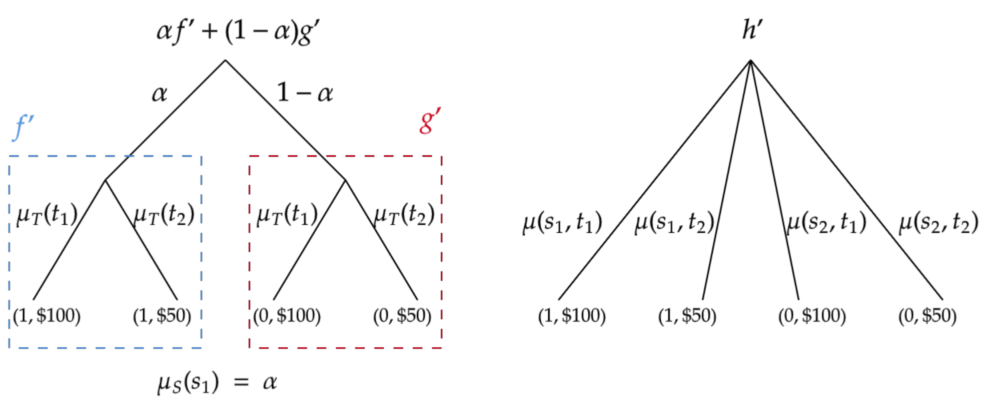

Figure 2: Intuition for Statistical Independence

In Figure 2, the DM's evaluation of h' is described by the right panel and her evaluation of $\alpha f' + (1 - \alpha)g'$ is described by the left panel. In particular, $\alpha f' + (1 - \alpha)g'$ is a two-stage lottery and its probability $\alpha, 1 - \alpha$ in the first stage is objective. In other words, our objective randomization device, which is statistically independent of the states of the world, can mimic the uncertainty in the S dimension. If μ is statistically independent, then the DM will be indifferent between $\alpha f' + (1 - \alpha)g'$ and h' since she is indifferent to whether the probability in the first stage is objective or subjective. This characterization will not immediately extend to pure Savage-style environments without objective randomization.

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6 Appendix

6.1 A Useful Lemma

Lemma 1. *(Uniqueness of Linear Function)*
Let R be a preference relation over $\mathcal{U} \subseteq \mathbb{R}^K$, where $\mathcal{U}$ is a convex set containing the origin $0^K \in \mathbb{R}^K$ and span$(\mathcal{U}) = \mathbb{R}^K$. Suppose there exists $\mu \in \mathbb{R}^K$ with $\sum_{k=1}^K \mu^k = 1$ such that, for any $r \in \mathbb{R}^K$,

$$V(r) = \sum_{k=1}^{K} \mu^k \cdot r^k = \mu r$$

defines a representation of R. Then μ is unique.

Proof. Suppose μ is not unique, there must exist $\mu' \neq \mu$ with $\sum_{k=1}^K \mu'^k = 1$ and R is represented by $V'(r) = \mu' r$.

We first claim that there exists $x \in \mathbb{R}^K$ such that $\mu x = 0$ and $\mu' x \neq 0$. To see this, suppose for any $x \in \mathbb{R}^K$, $\mu x = 0 \implies \mu' x = 0$. For any $i, j \in \{1, 2, ..., K\}, i \neq j$, consider $w_{ij} \in \mathbb{R}^K$ such that $w_{ij}^i = \mu^j, w_{ij}^j = -\mu^i, w_{ij}^k = 0$ for all $k \neq i, j$. Clearly, $\mu w_{ij} = 0$ so $\mu' w_{ij} = 0$, which implies $\mu'^i \mu^j = \mu'^j \mu^i$. If $\mu^i = 0$, we can choose some $\mu^j \neq 0$ (since $\sum_{k=1}^K \mu^k = 1$) to get $\mu'^i = 0$. A symmetric argument holds when $\mu'^i = 0$ and we thus have for any k, $\mu^k = 0 \iff \mu'^k = 0$. For nonzero μ^ks, since μ, μ' are normalized, $\mu'^i \mu^j = \mu'^j \mu^i$ for all $i, j \implies \mu^k = \mu'^k$. Together we obtain $\mu = \mu'$, a contradiction.

Let $\{b_1, b_2, ..., b_K\} \subset \mathcal{U}$ be a basis of $\mathbb{R}^K$ and its existence is guaranteed since span$(\mathcal{U}) = \mathbb{R}^K$. Say $x = \sum_{k=1}^K \alpha^k b_k$ and let $\overline{\alpha} = \sum_{k=1}^K \mid \alpha^k \mid$. Let $y = \frac{1}{K+1}(0^K + \sum_{k=1}^K b_k)$ and note that $y \in \mathcal{U}$ because $\mathcal{U}$ is convex and $0^K \in \mathcal{U}$. Consider $\epsilon = \frac{1}{\overline{\alpha}(K+1)+1}$ and we claim $(1-\epsilon)y + \epsilon x = \sum_{k=1}^K \beta^k b_k \in \mathcal{U}$. To see this, note that $\beta^k = \frac{\overline{\alpha}+\alpha^k}{\overline{\alpha}(K+1)+1}$ and $\beta^k \geq 0$ for all k by the definition of $\overline{\alpha}$. Furthermore, $\sum_{k=1}^K \beta^k = \frac{\overline{\alpha}K + \sum_{k=1}^K \alpha^k}{\overline{\alpha}(K+1)+1} \leq \frac{\overline{\alpha}K + \sum_{k=1}^K |\alpha^k|}{\overline{\alpha}(K+1)+1} = \frac{\overline{\alpha}(K+1)}{\overline{\alpha}(K+1)+1} < 1$. So $(1-\epsilon)y + \epsilon x \in \mathcal{U}$ because it is a convex combination of $\{b_1, b_2, ..., b_K\}$ and 0^K.

Finally, since $\mu((1-\epsilon)y + \epsilon x) = \mu(1-\epsilon)y + \epsilon \mu x = \mu((1-\epsilon)y + \epsilon 0^K)$, $(1-\epsilon)y + \epsilon 0^K$ is indifferent to $(1-\epsilon)y + \epsilon x$ according to μ. But $\mu'((1-\epsilon)y + \epsilon x) = \mu'(1-\epsilon)y + \epsilon \mu' x \neq \mu((1-\epsilon)y + \epsilon 0^K)$, so $(1-\epsilon)y + \epsilon 0^K$ is not indifferent to $(1-\epsilon)y + \epsilon x$ according to μ'. This yields a contradiction to μ, μ' both representing R. □

6.2 Proof of Theorem 1

Proof. The "if" direction is straightforward to verify and we prove the "only if" direction. The Reduction Axiom allows us to define a preference relation $\dot{\succeq}$ over $\phi(\triangle\mathcal{F})$ such that

$\phi(Q)\dot{\succeq}\phi(P)$ if and only if $Q \succeq P$. To save notations, we will use $\succeq$ over $\phi(\triangle\mathcal{F})$ instead of $\dot{\succeq}$.

We first restrict $\succeq$ on those constant acts $\phi(Q)$, that is, $\phi(Q)(s,t) = \phi(Q)(s',t')$ for all s,t (it is clear that this condition holds if and only if Q is a lottery over constant acts). The set of constant acts can be identified as $\triangle Z$. The $\succeq$ restricted on $\triangle Z$ satisfies weak order, continuity and independence axioms, which are required for the Expected Utility Theorem (e.g., see Kreps (1988)). So $\succeq$ over $q \in \triangle Z$ is represented by U, where $U(q) = \int_Z u(z)\, dq(z)$ and u is unique up to positive affine transformation. Fix this u. By the Non-triviality and Monotonicity Axioms, u cannot be constant valued and so as U.

Given an arbitrary $Q = \sum_{k=1}^K \alpha^k f^k \in \triangle\mathcal{F}$, we can define a mapping $\pi : \phi(\triangle\mathcal{F}) \to \mathbb{R}^{NM}$ such that the nmth coordinate of $\pi(\phi(Q))$, denoted as $\pi(\phi(Q))^{nm}$, is given by:

$$\begin{aligned}
\pi(\phi(Q))^{nm} &= \sum_{k=1}^{K} \alpha^k U\Big(\phi\big(f_S^k(s_n), f_T^k(t_m)\big)\Big) \\
&= U\left(\sum_{k=1}^{K} \alpha^k \phi\big(f_S^k(s_n), f_T^k(t_m)\big)\right) \\
&= U\big(\phi(Q)(s_n,t_m)\big)
\end{aligned}$$

The second equality follows since U is an expected utility. We claim the image of π, $\pi(\phi(\triangle\mathcal{F}))$ is a convex subset of $\mathbb{R}^{NM}$. Let $\pi(\phi(Q)), \pi(\phi(P)) \in \pi(\phi(\triangle\mathcal{F}))$, $\alpha \in (0,1)$ and $\phi(L) = \alpha\phi(P) + (1-\alpha)\phi(Q)$ ($\phi(L)$ exists because $\phi(\triangle\mathcal{F})$ is convex). It is sufficient to show $\alpha\pi(\phi(P)) + (1-\alpha)\pi(\phi(Q)) = \pi(\phi(L))$. For any n,m:

$$\begin{aligned}
&\big[\alpha\pi\big(\phi(P)\big) + (1-\alpha)\pi\big(\phi(Q)\big)\big]^{nm} \\
&= \alpha\pi(\phi(P))^{nm} + (1-\alpha)\pi(\phi(Q))^{nm} \\
&= \alpha\left(\sum_{k=1}^{K_P} \alpha_P^k U\Big(\phi\big(f_S^k(s_n), f_T^k(t_m)\big)\Big)\right) + (1-\alpha)\left(\sum_{j=1}^{J_Q} \alpha_Q^j U\Big(\phi\big(f_S^j(s_n), f_T^j(t_m)\big)\Big)\right) \\
&= U\left(\alpha\left[\sum_{k=1}^{K_P} \alpha_P^k \phi\left(f_S^k(s_n), f_T^k(t_m)\right)\right] + (1-\alpha)\left[\sum_{j=1}^{J_Q} \alpha_Q^j \phi\left(f_S^j(s_n), f_T^j(t_m)\right)\right]\right) \\
&= U\Big(\sum_{i=1}^{I_L} \alpha(L)^i \phi\big(f_S^i(s_n), f_T^i(t_m)\big)\Big) = \pi(\phi(L))^{nm}
\end{aligned}$$

We define a binary relation R over $\pi(\phi(\triangle\mathcal{F})$ such that $\pi(\phi(Q))R\pi(\phi(P))$ if and only if $\phi(Q) \succeq \phi(P)$, and let $\overline{R}$ be its strict part. To check R is well defined, it remains to verify that if $\pi(\phi(Q)) = \pi(\phi(P))$, then $\phi(Q) \sim \phi(P)$. Suppose $\phi(Q) \succ \phi(P)$, there must exist

s_n, t_m such that $\phi(Q)(s_n, t_m) \succ \phi(P)(s_n, t_m)$ since otherwise, it violates the Monotonicity Axiom. But this contradicts $U(\phi(Q)(s_n, t_m)) = \pi(\phi(Q))^{nm} = \pi(\phi(P))^{nm} = U(\phi(P)(s_n, t_m))$ since U represents lotteries over constant acts.

We show R defined on the mixture space $\pi(\phi(\triangle\mathcal{F}))$ satisfies all the conditions required for the Mixture Space Theorem. It is clear that R is a weak order since $\succeq$ is a weak order. We next show R satisfies independence, that is, for all $\phi(P), \phi(Q), \phi(L) \in \phi(\triangle\mathcal{F})$ and $\alpha \in (0,1)$, $\pi(\phi(P))R\pi(\phi(Q)) \iff \alpha\pi(\phi(P))+(1-\alpha)\pi(\phi(L))\ R\ \pi(\phi(Q))+(1-\alpha)\pi(\phi(L))$. Let $H^P = \alpha P + (1-\alpha)L, H^Q = \alpha Q + (1-\alpha)L$. Since we have proved $\pi(\phi(H^P)) = \alpha\pi(\phi(P))+(1-\alpha)\pi(\phi(L)), \pi(\phi(H^Q)) = \alpha\pi(\phi(Q))+(1-\alpha)\pi(\phi(L))$, we have $\alpha\pi(\phi(P))+(1-\alpha)\pi(\phi(L))\ R\ \pi(\phi(Q))+(1-\alpha)\pi(\phi(L)) \iff \pi(\phi(H^P))R\pi(\phi(H^Q)) \iff H^P \succeq H^Q \iff P \succeq Q$, where the last $\iff$ comes from the Independence Axiom. Finally, we show R satisfies continuity, that is, for any $\pi(\phi(P))\ \overline{R}\ \pi(\phi(Q))\ \overline{R}\ \pi(\phi(L))$, there exist $\alpha, \beta \in (0,1)$ such that $\alpha\pi(\phi(P)) + (1-\alpha)\pi(\phi(L))\ \overline{R}\ \pi(\phi(Q))\ \overline{R}\ \beta\pi(\phi(P)) + (1-\beta)\pi(\phi(L))$. By the Continuity Axiom, there exist $\alpha, \beta \in (0,1)$ and $\phi(H^\alpha) = \alpha\phi(P) + (1-\alpha)\phi(L), \phi(H^\beta) = \beta\phi(P) + (1-\beta)\phi(L)$ such that $H^\alpha \succ Q \succ H^\beta$. But since $\pi(\phi(H^\alpha)) = \alpha\pi(\phi(P)) + (1-\alpha)\pi(\phi(L)), \pi(H^\beta) = \beta\pi(\phi(P)) + (1-\beta)\pi(\phi(L))$, the desirable α, β are found.

Now we can apply the Mixture Space Theorem to R over the mixture space $\pi(\phi(\triangle\mathcal{F}))$, so there exists an linear function $V : \pi(\phi(\triangle\mathcal{F})) \to \mathbb{R}$ that represents R. Let $V(\pi(\phi(Q)) = \sum_{nm} w_{nm} \cdot \pi(\phi(Q))^{nm}$. By the Monotonicity Axiom, $w_{nm} \geq 0$ for all n, m. By the Non-triviality Axiom, there exist n, m such that $w_{nm} > 0$, so we could normalize the coefficients to $\mu(s_n, t_m) = w_{nm}/(\sum_{nm} w_{nm})$. Substituting the expression of $\pi(\phi(Q))^{nm}$ yields the representation (1).

Finally, we prove the marginal belief over S is unique (proof for T is symmetric). Fix some $z_T \in Z_T$ such that there exist $f, g \in \mathcal{F} \mid z_S$ with $f \succ g$, where $\mathcal{F} \mid z_T := \{f \in \mathcal{F} : f_T(t) = z_T \text{ for all } t \in T\}$. The existence of z_T is guaranteed by the Non-triviality Axiom. Consider the restriction of $\succeq$ over $\triangle(\mathcal{F} \mid z_T) \subseteq \triangle\mathcal{F}$, which is the same domain used by Anscombe and Aumann (1963). By their version of the SEU theorem, the subjective belief over S, which is the marginal belief μ_S in our setting, is unique. □

6.3 Proof of Theorem 2

Proof. The "if" direction is easy to check and we prove the "only if". In the proof of the "only if" direction of Theorem 1, we show $\succeq$ over $q \in \triangle Z$ is represented by U, where $U(q) = \int_Z u(z)\, dq(z)$ and u is unique up to positive affine transformation. The Additive Separability Axiom implies Axiom A7* in Theorem 2 of Chapter 6 in Fishburn (1982) and hence there exist u_S, u_T such that $u(z^j) = u_S(z^j_S) + u_T(z^j_T)$ and u_S, u_T are unique up to joint positive affine transformation. The remainder of the proof is the same as the proof of Theorem 1.

It remains to prove the statement about μ. Given the additively separable u, we can rewrite V as follows:

$$\begin{aligned} V(Q) &= \sum_{k=1}^{K} \alpha^k \left[\int_{S\times T} U\Big(\phi\big(f_S^k(s), f_T^k(t)\big)\Big) d\mu(s,t) \right] \\ &= \sum_{k=1}^{K} \alpha^k \left[\int_{S\times T} \left(\sum_{z_S\in Z_S} f_S^k(s)(z_S)\cdot u_S(z_S) + \sum_{z_T\in Z_T} f_T^k(t)(z_T)\cdot u_T(z_T) \right) d\mu(s,t) \right] \\ &= \sum_{k=1}^{K} \alpha^k \left[\int_{S} \left(\sum_{z_S\in Z_S} f_S^k(s)(z_S)\cdot u_S(z_S) \right) d\mu_S(s) + \int_{T} \left(\sum_{z_T\in Z_T} f_T^k(t)(z_T)\cdot u_T(z_T) \right) d\mu_T(t) \right] \end{aligned}$$

Therefore, if $\mu'_S = \mu_S, \mu'_T = \mu_T$, (U, μ') also represents $\succeq$. □

6.4 Proof of Theorem 3

Proof. The equivalence between 1 and 2 is established by Theorem 2. We first show 2 $\implies$ 3. Let $f_T, f'_T \in \mathcal{F}_T$ be arbitrary. Let $\succeq$ have an SEU representation (u, μ) and u be additively separable. Note that under the SEU with additive separability, $f_T \succeq_{f_S} f'_T$ if and only if $(f_S, f_T) \succeq (f'_S, f'_T)$ (with $f'_S = f_S$) if and only if $V(f) \geq V(f')$, that is,

$$\begin{aligned} &\sum_m \mu_T(t_m)\cdot\Big(\sum_{z_T\in Z_T} f_T(t_n)(z_T)\cdot u_T(z_T)\Big) + \sum_n \mu_S(s_n)\cdot\Big(\sum_{z_S\in Z_S} f_S(s_n)(z_S)\cdot u_S(z_S)\Big) \geq \\ &\sum_m \mu_T(t_m)\cdot\Big(\sum_{z_T\in Z_T} f'_T(t_n)(z_T)\cdot u_T(z_T)\Big) + \sum_n \mu_S(s_n)\cdot\Big(\sum_{z_S\in Z_S} f'_S(s_n)(z_S)\cdot u_S(z_S)\Big) \\ &\iff \sum_m \mu_T(t_m)\cdot\Big(\sum_{z_T\in Z_T} f_T(t_n)(z_T)\cdot u_T(z_T)\Big) \geq \sum_m \mu_T(t_m)\cdot\Big(\sum_{z_T\in Z_T} f'_T(t_n)(z_T)\cdot u_T(z_T)\Big) \end{aligned}$$

The first inequality comes from the additive separability and the representation V. The last line follows since the second term on both sides of the inequality only depends on f_S, f'_S and $f_S = f'_S$. But clearly, the last inequality does not depend on the choice of f_S and thus statement 3 follows.

Now we prove 3 $\implies$ 2. We first show that 3 $\implies$ Axiom A7* in Theorem 2 of Chapter 6 in Fishburn (1982), which is translated using our terminology as follows:

Axiom 7*: Let $P, Q \in \triangle\mathcal{F}$ be lotteries over constant acts so $\phi(P), \phi(Q) \in \triangle Z$. If $\phi(P)(z), \phi(Q)(z) \in \{0, \frac{1}{2}, 1\}$ for all $z \in Z$ and $\phi(P)_S = \phi(Q)_S, \phi(P)_T = \phi(Q)_T$, then $P \sim Q$.

We only need to prove for the following case: for any $z_S, z'_S \in Z_S, z_T, z'_T \in Z_T$, $P =$

$\frac{1}{2}(z_S, z_T) + \frac{1}{2}(z'_S, z'_T) \sim Q = \frac{1}{2}(z'_S, z_T) + \frac{1}{2}(z_S, z'_T)$. This is because for the rest of the cases, the conditions in Axiom 7* simply imply $P = Q$. To prove this case, note that the statement $3 \implies (z_S, z_T) \sim (z'_S, z_T)$ and $(z'_S, z'_T) \sim (z_S, z'_T)$. Then the SEU representation (1) immediately implies $P \sim Q$.

By Theorem 2 of Chapter 6 in Fishburn (1982), Axiom 7* implies the additively separable u as desired.

The proof for statement 4 is symmetric to the proof for statement 3. □

6.5 Proof of Theorem 4

Proof. Since $\succeq$ over $\triangle\mathcal{F}$ has an SEU representation (U, μ), $\succeq$ over $\triangle\mathcal{F}^{st}$ also has an SEU representation. $f = ((p_S, p'_S)^s, (p_T, p'_T)^t) \in \triangle\mathcal{F}^{st}$ is evaluated by

$$\begin{aligned} V(f) =& \mu(s,t)U\Big(\phi(p_S, p_T)\Big) + \mu(S \setminus s, t)U\Big(\phi(p'_S, p_T)\Big) \\ &+ \mu(s, T \setminus t)U\Big(\phi(p_S, p'_T)\Big) + \mu(S \setminus s, T \setminus t)U\Big(\phi(p'_S, p'_T)\Big) \end{aligned}$$

Following a similar strategy as in the proof of Theorem 1, we can define a preference relation R over $\pi(\phi(\triangle\mathcal{F}^{st}))$ such that $\pi(\phi(Q))R\pi(\phi(P))$ if and only if $Q \succeq P$ for any $P, Q \in \triangle\mathcal{F}^{st}$. Note that each element $\pi(\phi(Q)) \in \pi(\phi(\triangle\mathcal{F}^{st}))$ can be identified as an element in $\mathbb{R}^4$ because by construction $\pi^{nm}(\phi(Q)) = \pi^{ij}(\phi(Q))$ if $s_n, s_i \in S \setminus s, t_m, t_j \in T \setminus t$. In particular, for the f above, $\pi(\phi(f))$ can be identified by

$$\bigg(U\Big(\phi(p_S, p_T)\Big), U\Big(\phi(p'_S, p_T)\Big), U\Big(\phi(p_S, p'_T)\Big), U\Big(\phi(p'_S, p'_T)\Big)\bigg) \in \mathbb{R}^4$$

Thus V is a linear function representing a preference relation R over $\pi(\phi(\triangle\mathcal{F}^{st}))$, which is identified as a subset of $\mathbb{R}^4$. If we can show the linear representation V above is unique then we are done, since $\mu(s,t)$ is then unique for arbitrary s, t. By Lemma 1, it is sufficient to show that $\pi(\phi(\triangle\mathcal{F}^{st}))$ is convex, contains the zero vector, and spans $\mathbb{R}$. The convexity follows by construction. Because U in the SEU representation is unique up to positive affine transformation, we can choose a U such that $\mathcal{U}$ contains the zero vector in $\mathbb{R}^4$ by considering a constant act and shifting it uniformly to the origin.

It thus remains to prove $\pi(\phi(\triangle\mathcal{F}^{st}))$ spans $\mathbb{R}^4$. Let $\mathcal{U} =$

$$\left\{\bigg(U\Big(\phi(p_S, p_T)\Big), U\Big(\phi(p'_S, p_T)\Big), U\Big(\phi(p_S, p'_T)\Big), U\Big(\phi(p'_S, p'_T)\Big)\bigg) \in \mathbb{R}^4 : p_S, p'_S \in \triangle Z_S, p_T, p'_T \in \triangle Z_T\right\}$$

It is sufficient to prove $span(\mathcal{U}) = \mathbb{R}^4$ because $\pi(\phi(\triangle\mathcal{F}^{st}))$ is the convex hull of $\mathcal{U}$.

To save notations, we will use a, b to denote elements in $\triangle Z_S$ and x, y to denote elements in $\triangle Z_T$ and will write $U(\phi(p_S, p_T))$ as $U(a,x)$ for $a = p_S, x = p_T$. These notations are used exclusive for this proof. We prove by contradiction. Suppose $span(\mathcal{U}) \neq \mathbb{R}^4$, then there exists $A = (A^1, A^2, A^3, A^4) \in \mathbb{R}^4$ such that $A \neq 0^4$ and $A \cdot r = \sum_{k=1}^4 A^k \cdot r^k = 0$ for all $r \in \mathcal{U}$.

By the Non-triviality Axiom, there exist $a, b \in \triangle Z_S, x \in \triangle Z_T$ such that $U(a,x) > U(b,x)$. So $U(a,x), U(b,x)$ cannot both equal to zero and say $U(a,x) \neq 0$. Consider $r = (U(a,x), U(a,x), U(a,x), U(a,x))$, then $A \cdot r = 0 \implies \sum_{k=1}^4 A^k = 0$.

We claim there exist $a, b \in \triangle Z_S, x, y \in \triangle Z_T$ such that $U(a,x) + U(b,y) \neq U(b,x) + U(a,y)$. To see this, suppose it is not true, then Axiom 7* in Theorem 2 of Chapter 6 in Fishburn (1982) holds. But Axiom 7* is equivalent to the Additive Separability Axiom in our context. Let $r_1 = (U(a,x), U(a,y), U(b,x), U(b,y)), r_2 = (U(b,y), U(b,x), U(a,y), U(a,x))$. Then $A \cdot r_1 + A \cdot r_2 = 0$ yields

$$\begin{aligned}
&(U(a,x) + U(b,y))(A^1 + A^4) + (U(b,x) + U(a,y))(A^2 + A^3) = 0 \\
\implies &\Big(U(a,x) + U(b,y) - (U(b,x) + U(a,y))\Big)(A^1 + A^4) = 0 \quad \text{since } \sum_{k=1}^4 A^k = 0 \\
\implies &A^1 + A^4 = 0 \implies A^2 + A^3 = 0
\end{aligned}$$

So there exist $a, b \in \triangle Z_S, x, y \in \triangle Z_T$ such that $U(a,x) + U(b,y) \neq U(b,x) + U(a,y)$, say $U(a,x) + U(b,y) > U(b,x) + U(a,y)$. Let $r_1 = (U(a,x), U(a,y), U(b,x), U(b,y)), r_2 = (U(b,y), U(b,x), U(a,y), U(a,x))$. Then $A \cdot r_1 + A \cdot r_2 = 0$ yields

$$\begin{aligned}
&(U(a,x) + U(b,y))(A^1 + A^4) + (U(b,x) + U(a,y))(A^2 + A^3) = 0 \\
\implies &\Big(U(a,x) + U(b,y) - (U(b,x) + U(a,y))\Big)(A^1 + A^4) = 0 \quad \text{since } \sum_{k=1}^4 A^k = 0 \\
\implies &A^1 + A^4 = 0 \implies A^2 + A^3 = 0
\end{aligned}$$

Let $r_3 = (U(a,x), U(a,y), U(a,x), U(a,y))$, where $U(a,x) \neq U(a,y)$. The existence of such $a \in \triangle Z_S, x, y \in \triangle Z_T$ is again guaranteed by the Non-triviality Axiom. Then $A \cdot r_3 = 0$ implies

$$\begin{aligned}
&U(a,x)(A^1 + A^3) + U(a,y)(A^2 + A^4) = 0 \\
\implies &U(a,x)(A^1 - A^2) + U(a,y)(A^2 - A^1) = 0 \quad \text{since } A^1 + A^4 = 0, A^2 + A^3 = 0 \\
\implies &(U(a,x) - U(a,y))(A^1 - A^2) = 0 \implies A^1 = A^2
\end{aligned}$$

It follows $A = (A^1, A^1, -A^1, -A^1) \neq 0^4$. Let $r_4 = (U(a,x), U(a,x), U(b,x), U(b,x))$ for arbitrary $a, b \in \triangle Z_S, x \in \triangle Z_T$. Then $A \cdot r_4 = 0$ implies

$$\begin{aligned} & 2A^1 U(a,x) - 2A^1 U(b,x) = 0 \\ \Longrightarrow\; & U(a,x) = U(b,x) \quad \text{since } A^1 \neq 0 \end{aligned}$$

But $a, x \sim b, x$ for all $a, b \in \triangle Z_S, x \in \triangle Z_T$ is contradicting to the Non-triviality Axiom. $\square$

6.6 Proof of Theorem 5

Proof. The "if" direction is easy to verify. We prove the "only if" direction. The first part of the Monetary Payoffs Axiom is stronger than the Reduction Axiom so we can obtain an SEU representation. Let u be its utility index. For each $z \in \mathbb{R}$, define $v(z) := u(x, y)$ for some $x, y \in \mathbb{R}$ such that $x + y = z$. This is well defined because by the Monetary Payoffs Axiom, $x + y = x' + y' \implies \phi^*(\delta_{(x,y)}) = \phi^*(\delta_{(x',y')}) \implies \delta_{(x,y)} \sim \delta_{(x',y')} \implies u(x, y) = u(x', y')$. The second part of the axiom immediately implies v is strictly increasing. $\square$

6.7 Proof of Theorem 6

Proof. We first show $1 \implies 2$. Suppose $\succeq$ has an SEU representation (u, μ) and μ is statistically independent. Fix $s \in S$ and consider $f, g, f', g' \in \mathcal{F} \mid S$ such that $f_T = g_T, f'_T = g'_T$. Suppose $f \not\sim g$. Note that by the representation, $h = f \diamond_s g \sim \alpha f + (1-\alpha) g$ is equivalent to any of the following equations:

$$\sum_{n,m} \mu(s_n, t_m) U\Big(\phi\big(h_S(s_n), h_T(t_m)\big)\Big) = \begin{array}{l} \alpha \sum_{n,m} \mu(s_n, t_m) U\Big(\phi\big(f_S(s_n), f_T(t_m)\big)\Big) \\ \quad + (1-\alpha) \sum_{n,m} \mu(s_n, t_m) U\Big(\phi\big(g_S(s_n), g_T(t_m)\big)\Big) \end{array} \tag{4}$$

$$\begin{array}{l} \sum_m \mu(s, t_m) U\Big(\phi\big(f_S(s), h_T(t_m)\big)\Big) \\ \quad + \sum_{s_n \neq s, m} \mu(s_n, t_m) U\Big(\phi\big(g_S(s), h_T(t_m)\big)\Big) \end{array} = \begin{array}{l} \alpha \sum_{n,m} \mu(s_n, t_m) U\Big(\phi\big(f_S(s), h_T(t_m)\big)\Big) \\ \quad + (1-\alpha) \sum_{n,m} \mu(s_n, t_m) U\Big(\phi\big(g_S(s), h_T(t_m)\big)\Big) \end{array} \tag{5}$$

$$\begin{array}{l} \mu_S(s) \sum_m \mu_T(t_m) U\Big(\phi\big(f_S(s), h_T(t_m)\big)\Big) \\ \quad + (1-\mu_S(s)) \sum_m \mu_T(t_m) U\Big(\phi\big(g_S(s), h_T(t_m)\big)\Big) \end{array} = \begin{array}{l} \alpha \sum_m \mu_T(t_m) U\Big(\phi\big(f_S(s), h_T(t_m)\big)\Big) \\ \quad + (1-\alpha) \sum_m \mu_T(t_m) U\Big(\phi\big(g_S(s), h_T(t_m)\big)\Big) \end{array} \tag{6}$$

$$(\mu_S(s) - \alpha) \left[\sum_m \mu(s, t_m) U\Big(\phi\big(f_S(s), h_T(t_m)\big)\Big) - \sum_m \mu(s, t_m) U\Big(\phi\big(g_S(s), h_T(t_m)\big)\Big) \right] = 0 \tag{7}$$

To explain, (4) is equivalent to (5) because $f, g \in \mathcal{F} \mid S$, the definition of $\diamond_s$ and $f_T = g_T$. Next, (5) is equivalent (6) because μ is statistically independent. Rearranging the terms in (6) yields (7). From (7), it follows $\mu_S(s) = \alpha$ because $f \not\sim g$ implies the second term in the square bracket being nonzero. Following almost exactly the same steps as above, $f' \diamond_s g' \sim \alpha f' + (1-\alpha) g'$ holds if and only if

$$(\mu_S(s) - \alpha)\left[\sum_m \mu(s, t_m) U\Big(\phi\big(f'_S(s), f'_T(t_m)\big)\Big) - \sum_m \mu(s, t_m) U\Big(\phi\big(g'_S(s), f'_T(t_m)\big)\Big)\right] = 0$$

But this equality holds because we have proved $\mu_S(s) = \alpha$.

Next we prove $2 \implies 1$. We only prove for the case when the belief μ is unique in the SEU representation This is because otherwise, by Theorem 4, μ is identified only up to marginals and we can construct a statistically independent μ by the product measure of its unique marginals. Let $s \in S, t \in T$ be arbitrary. Consider constant acts $f = (p_S, p_T), g = (p'_S, p_T)$ such that $f \not\sim g$ and $f \diamond_s g \sim \alpha f + (1-\alpha) g$. By the SEU representation,

$$\begin{aligned}
V(f \diamond_s g) &= \mu_S(s) U(\phi(p_S, p_T)) + (1 - \mu_S(s)) U(\phi(p'_S, p_T)) \\
V(\alpha f + (1-\alpha) g) &= \alpha U(\phi(p_S, p_T)) + (1-\alpha) U(\phi(p'_S, p_T)) \\
\text{so } f \diamond_s g \sim \alpha f + (1-\alpha) g &\iff V(f \diamond_s g) = V(\alpha f + (1-\alpha) g) \\
&\iff (\mu_S(s) - \alpha)\Big(U(\phi(p_S, p_T)) - U(\phi(p'_S, p_T))\Big) = 0
\end{aligned}$$

Note that $f \not\sim g \iff U(\phi(p_S, p_T)) \neq U(\phi(p'_S, p_T))$. It follows $f \diamond_s g \sim \alpha f + (1-\alpha) g \iff \alpha = \mu_S(s)$.

Now consider $f', g' \in \mathcal{F} \mid S$ such that $f'_S = p_S, g'_S = p'_S$ and $f'_T(t) = g'_T(t) = p_T$, $f'_T(t') = g'_T(t') = p'_T$ for all $t' \neq t$. Because $\succeq$ satisfies the Statistical Independence S Axiom, we have $f' \diamond_s g' \sim \alpha f' + (1-\alpha) g'$. By the SEU representation,

$$\begin{aligned}
V(f' \diamond_s g') =& \mu(s, t) U\Big(\phi(p_S, p_T)\Big) + \mu(S \setminus s, t) U\Big(\phi(p'_S, p_T)\Big) \\
&+ \mu(s, T \setminus t) U\Big(\phi(p_S, p'_T)\Big) + \mu(S \setminus s, T \setminus t) U\Big(\phi(p'_S, p'_T)\Big) \\
V(\alpha f' + (1-\alpha) g') =& \alpha\Big[\mu_T(t) U\Big(\phi(p_S, p_T)\Big) + \mu_T(T \setminus t) U\Big(\phi(p_S, p'_T)\Big)\Big] \\
&+ (1-\alpha)\Big[\mu_T(t) U\Big(\phi(p'_S, p_T)\Big) + \mu_T(T \setminus t) U\Big(\phi(p'_S, p'_T)\Big)\Big]
\end{aligned}$$

So we have

$$f' \diamond_s g' \sim \alpha f' + (1-\alpha) g' \iff V(f' \diamond_s g') = V(\alpha f' + (1-\alpha) g')$$

Substituting $\alpha = \mu_S(s)$ yields $f' \diamond_s g' \sim \alpha f' + (1-\alpha) g' \iff$

$$0 = \Big[\mu(s,t) - \mu_S(s)\mu_T(t)\Big] U(\phi(p_S, p_T)) + \Big[\mu(s, T\setminus t) - \mu_S(s)\mu_T(T\setminus t)\Big] U(\phi(p_S, p'_T)) + \Big[\mu(S\setminus s, t) - \mu_S(S\setminus s)\mu_T(t)\Big] U(\phi(p'_S, p_T)) + \Big[\mu(S\setminus s, T\setminus t) - \mu_S(S\setminus s)\mu_T(T\setminus t)\Big] U(\phi(p'_S, p'_T))$$

Note that the above equation holds for all $p_S, p'_S \in \triangle Z_S, p_T, p'_T \in \triangle Z_T$. As one step in the proof of Theorem 4, we have shown the span of the set

$$\left\{\left(U\Big(\phi(p_S,p_T)\Big), U\Big(\phi(p'_S,p_T)\Big), U\Big(\phi(p_S,p'_T)\Big), U\Big(\phi(p'_S,p'_T)\Big)\right) \in \mathbb{R}^4 : p_S, p'_S \in \triangle Z_S, p_T, p'_T \in \triangle Z_T\right\}$$

is $\mathbb{R}^4$. It follows the four terms in the square brackets of the above equation must all be zero. Therefore, $\mu(E_S, E_T) = \mu_S(E_S) \cdot \mu_T(E_T)$ for $E_S \in \{s, S\setminus s\}, E_T \in \{t, T\setminus t\}$. Since the choice of s, t are arbitrary, we conclude $\mu = \mu_S \cdot \mu_T$.

Finally, the proof for $1 \iff 3$ is symmetric. $\square$